

Titanic Dataset EDA Summary

1. Data Loading and Initial Inspection

- Imported necessary libraries (pandas, numpy, matplotlib, seaborn)
- Loaded Titanic dataset from 'train.csv'
- Initial data exploration using df.head() and df.info()

2. Missing Value Analysis

- Visualized missing values using barplot

Key findings:

- Cabin: Highest missing values
- Age: Moderate missing values
- Embarked: Few missing values

3. Numerical Variables Analysis

Created distribution plots for numerical features:

- PassengerId
- Survived
- Pclass
- Age
- SibSp
- Parch
- Fare

Generated correlation matrix heatmap showing relationships between numerical variables

4. Categorical Variables Analysis

Analyzed distributions of:

- Sex
- Cabin
- Embarked
- Name-derived features
- Used countplots to visualize categorical variable distributions

5. Survival Analysis

Overall survival rate: ~38%

Survival by demographics:

- Sex: Females had higher survival rate
- Class: First class passengers had better survival odds
- Age: Children had higher survival rate

5. Feature Relationships

Created scatter plots showing:

- Age vs. Fare by Survival
- Class vs. Age distribution
- Fare distribution across classes
- Generated pair plots for numerical variables

6. Advanced Visualizations

Combined histogram and scatter plots showing:

- Age distribution by survival
- Fare distribution by survival
- Class distribution by survival
- Multivariate relationships between Age, Fare, Class, and Survival

Key Insights

- Gender was the strongest predictor of survival
- Higher passenger class correlated with better survival rates
- Children had better survival odds
- Higher fare paying passengers had better survival rates
- Family size had a moderate impact on survival
- Port of embarkation showed some correlation with survival

This EDA provides a comprehensive understanding of the Titanic dataset and the factors that influenced passenger survival, which can be used for feature selection and modeling